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8.6.3 Factors Affecting the Rate of Enzyme Reactions

Enzymes are sensitive to environmental condition. It could be affected by temperature, pH, substrate concentration and enzyme concentration.

The rate of enzyme reaction is measured by the amount of substrate changed or amount of product formed, during a period of time.

8.6.4 Temperature

Heating increases molecular motion. Thus the molecules of the substrate and enzyme move more quickly resulting in a greater probability of occurence of the reaction. The temperature that promotes maximum activity is referred to as optimum temperature (Figure 8.31a).

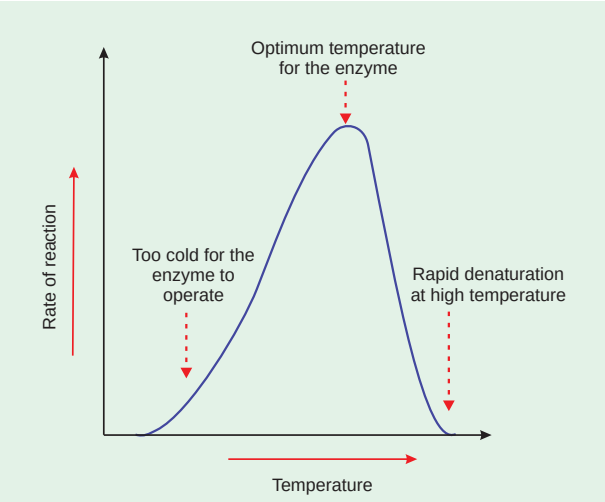

Figure 8.31(a): Temperature

8.6.5 pH

The optimum pH is that at which the maximum rate of reaction occurs. Thus the pH change leads to an alteration of enzyme shape, including the active site. If extremes of pH are encountered by an enzyme, then it will be denatured (Figure 8.31b).

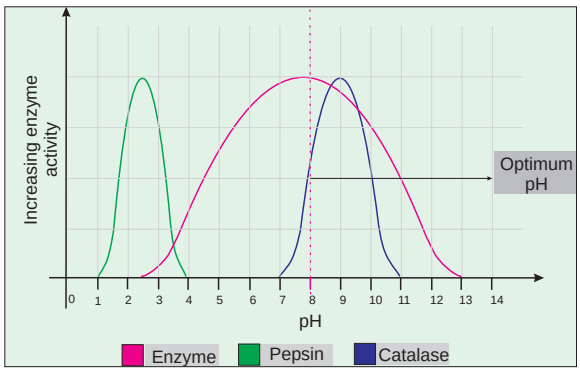

Figure 8.31(b): pH

8.6.6 Substrate Concentration

For a given enzyme concentration, the rate of an enzyme reaction increases with increasing substrate concentration (Figure 8.32).

8.6.7 Enzyme Concentration

The rate of reaction is directly proportional to the enzyme concentration.

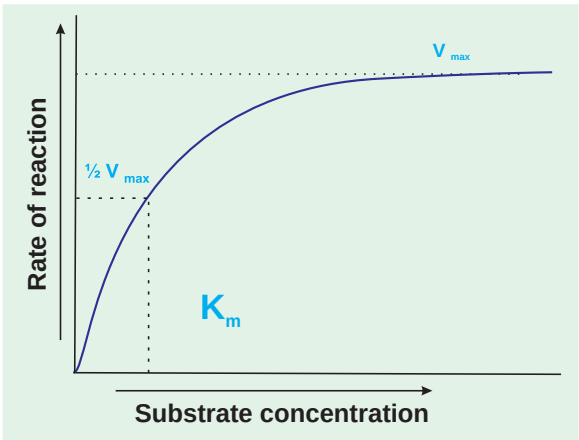

Figure 8.32: Rate of enzyme reaction

8.6.8 Introducing the Michaelis-Menton Constant (Km) and Its Significance

When the initial rate of reaction of an enzyme is measured over a range of substrate concentrations (with a fixed amount of enzyme) and the results plotted on a graph. With increasing substrate concentration, the velocity increases – rapidly at lower substrate concentration.